

Signed Number :-

1 0 1 0 1 1 0 0

2^5 Complement

0 1 0 1 0 1 0 0

↑ invert

0 0 0 0

0 0 1 $\rightarrow$ 1

-4 to 3

0 1 0 $\rightarrow$ 2

0 0 0 $\rightarrow$ 3

$-(2^{3-1})$ to (2^{3-1})

1 0 0 $\rightarrow$ -4

-4 to +3

1 0 1 $\rightarrow$ -3

1 1 0 $\rightarrow$ -2

1 1 1 $\rightarrow$ -1

Binary to gray .

1 0 1 1 0 (2)

→
xor

1 1 1 0 1

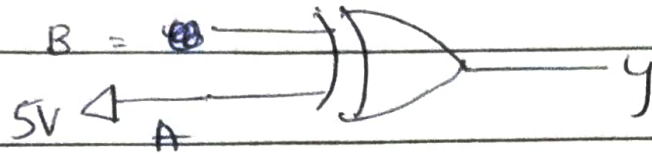
(Gray)

↓
xor

(Binary)

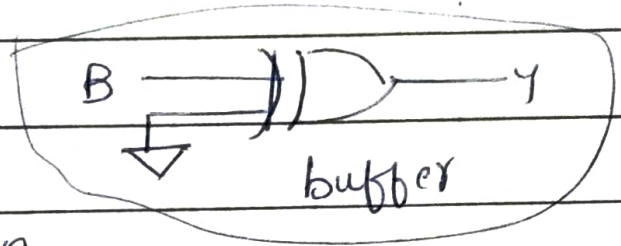
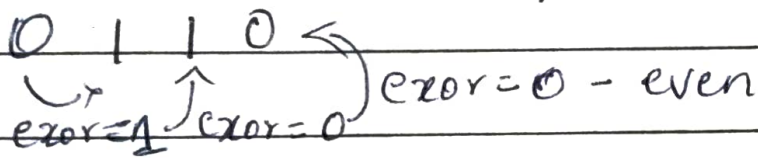
Exor to NOT gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



A = 1 B = 0 y = 1
B = 1 y = 0

detect odd parity

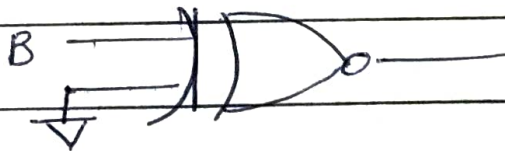


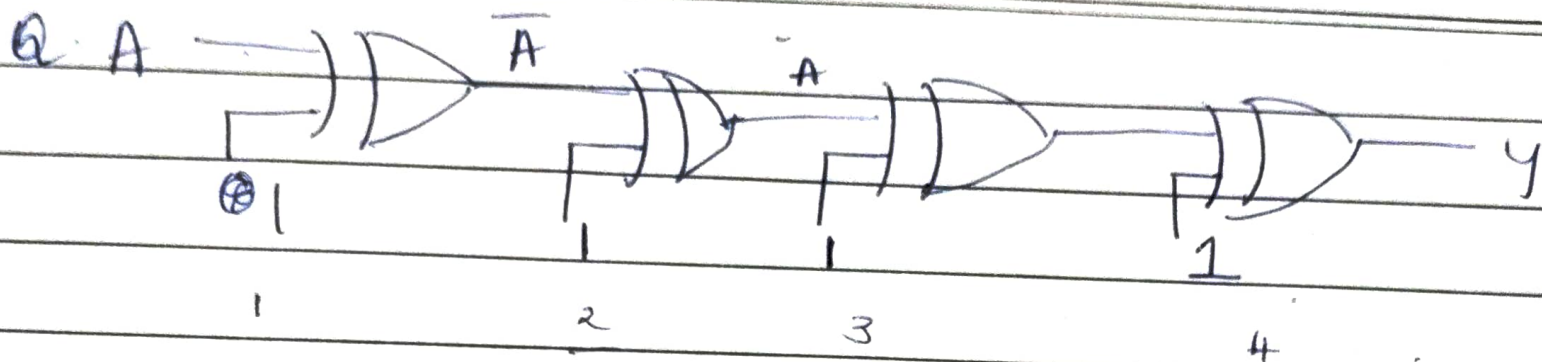
Exnor :-

detect even parity

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

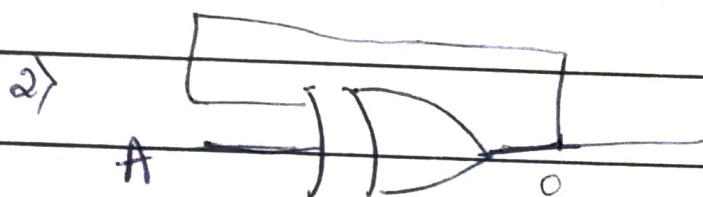
Exnor as not





After 500 xor $y = ?$

even as 2 $y = A$ (as same as xor2)



$$A \oplus A$$

$$A \oplus 1 = \bar{A}$$

$$0 \oplus 0$$

$$0$$

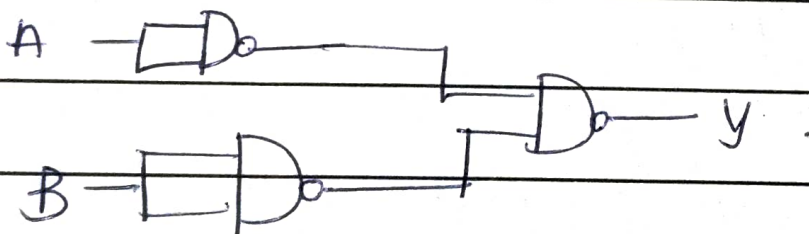
$$A \oplus \bar{A} = 1$$

$$1 \oplus 1$$

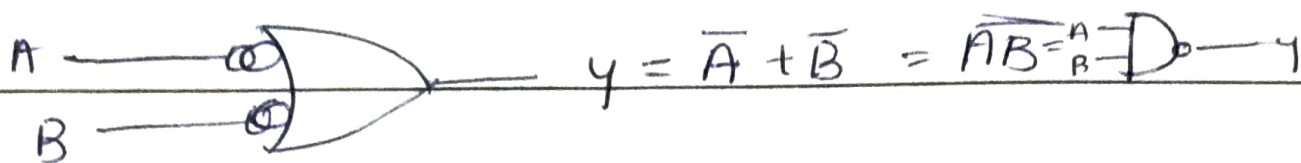
$$0$$

$$A \oplus 1 = \bar{A}$$

NAND to OR



Double complement 1st step.



$$\overline{A} \cdot \overline{A} = \overline{A} \quad A \cdot \overline{A} \cdot A = 0$$

$$A \cdot \overline{A} = 0 \quad A + B = (A + B)(A + C)$$

$$1 \cdot \overline{A} = \overline{A} \quad \overline{A} \cdot B + B = B + \overline{A} \cdot B$$

$$1 \cdot A = A \quad (B + \overline{A})(B + \overline{B}) = B + \overline{A}$$

$$AB + \overline{A}C + \overline{B}C = AB + \overline{A}C$$

$$AB + \overline{A}C + (A + \overline{A})(B + \overline{B})$$

!

$$P(P + \text{Anything}) = P$$

$$\text{Ex: } A(A + AB + BC + CD + CE + DE) = A$$

Min term - SOP defined when o/p = 1

Max term = POS